



Niels Henrik Abel Mathematics Competition: 2018–2019 Finals – *Solutions*

March 5, 2019



Problem 1. We can think of earned points as the sum of two types of points: row points and column points. When you mark a row with i marks already present, you receive i row points. In total, you then receive $0 + 1 + 2 + \dots + (n - 1) = n(n - 1)/2$ row points from this row. From all n rows, you get $n^2 (n - 1)/2$ row points. Similarly, you get $n^2 (n - 1)/2$ column points, and thus a total of $n^2 (n - 1)$ points in all, regardless of the order in which you place the crosses.

Problem 2. First, we note that for $m = 1$, $mn - 1 = n - 1 \mid n^3 - 1$ (because $n^3 - 1 = (n - 1)(n^2 + n + 1)$). And if $n = 1$, then $n^3 - 1 = 0$, and every number is divisible by 0. Thus, every pair of numbers (m, n) where $m = 1$ or $n = 1$ satisfies the conditions of the problem.

When examining other possibilities, we can therefore assume $m > 1$ and $n >$

1. If $m = n^2$, the condition is satisfied (it becomes $n^3 - 1 \mid n^3 - 1$), and if $n = m^2$, it is also satisfied (it becomes $m^3 - 1 \mid m^6 - 1$, which is true because $m^6 - 1 = (m^3 - 1)(m^3 + 1)$).

The given condition yields, in turn:

$$\begin{aligned} mn - 1 \mid n^3 - 1, & \quad mn - 1 \mid m(n^3 - 1) = (mn - 1)n^2 + (n^2 - m), \\ mn - 1 \mid n^2 - m, & \quad mn - 1 \mid m(n^2 - m) = (mn - 1)n + (n - m^2), \\ mn - 1 \mid n - m^2, & \quad mn - 1 \mid m(n - m^2) = (mn - 1) + (1 - m^3), \\ mn - 1 \mid m^3 - 1. & \end{aligned}$$

Since we now assume that $n > 1$, $mn - 1 \mid n^3 - 1$ implies that $mn - 1 \leq n^3 - 1$, so $m \leq n^2$. Similarly, the assumption $m > 1$, together with $mn - 1 \mid m^3 - 1$, implies that $n \leq m^2$.

Now suppose that $m < n^2$. Then $mn - 1 \mid n^2 - m$ gives us the inequality $mn - 1 \leq n^2 - m$, that is, $mn \leq n^2 - m + 1 < n^2$, so $m < n$. If $n < m^2$ as well, we similarly obtain $n < m$, which is a contradiction. Therefore, there are no more possibilities than those we have already found.



Problem 3.

a. If the radii of the three circles are a , b , and c , then the perimeter of the triangle with vertices at the centers of the circles is equal to $2a + 2b + 2c$, so $a + b + c = \frac{1}{2}$, while the sum of areas is $\pi(a^2 + b^2 + c^2)$. We know that

$$\sqrt{\frac{a^2 + b^2 + c^2}{3}} \geq \frac{a + b + c}{3} \quad (\text{AM-QM})$$

with equality if and only if $a = b = c$ (in which case it equals $\frac{1}{6}$). This gives $\pi(a^2 + b^2 + c^2) \geq \frac{\pi}{12}(a + b + c)^2$, with equality when all three circles have the same radius.

b. By substituting $x = 1$ and $y = 2019$, we see that $f(1) = 1$. Next, we substitute $y = 1$, which gives $f(2019/x) = f(x)$. Now we see that the original functional equation yields $f(x)f(y) = f(xy)$. In particular, $f(x)^2 = f(x^2)$, so $f(x^2) > 0$, which means

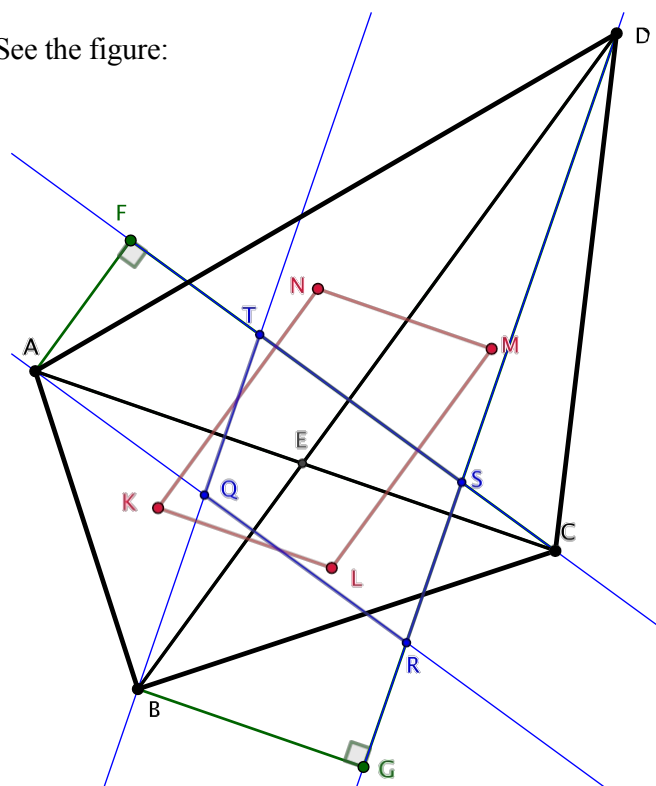
$f(x) > 0$ for $x > 0$. Furthermore, we now have $f(x)^2 = f(x)f(2019/x) = f(x(2019/x)) =$

$f(2019) = 1$, so $f(x) = \pm 1$. Thus, $f(x) = 1$ for all $x > 0$. If $x < 0$

and $y > 0$, then $f(x) = f(x)f(y) = f(xy)$. This implies that $f(x)$ has the same value for all $x < 0$. We have two possibilities: $f(x) = 1$ for all $x \neq 0$, or $f(x) = 1$ for $x > 0$ and $f(x) = -1$ for $x < 0$. Both of these satisfy the requirements of the problem.



Problem 4. See the figure:



We start with the quadrilateral $LMNK$ (red in the figure): It is known that the height from a base to the centroid of a triangle is one-third of the height to the vertex. It follows that $MN \parallel AC \parallel KL$ and $LM \parallel BD \parallel KN$, so $LMNK$ is a parallelogram, and $KN = \frac{1}{3}BD$ while $KL = \frac{1}{3}AC$. In particular, $KN : KL = BD : AC$.

The orthocenters Q, R, S , and T (the vertices of the blue square) are where the normals from A, B, C and D meet at the diagonals AC and BD , and the sides of the quadrilateral are segments of these normals. In particular, $RS \perp AC$ and $QT \perp AC$, so $RS \parallel QT$. Similarly, $QR \perp BD$ and $ST \perp BD$, so $QR \parallel ST$. Thus, $QRST$ is also a parallelogram.

Since the sides of $LMNK$ are perpendicular to the sides of $QRST$, $\angle QRS = \angle NKL$ and $\angle TQR = \angle MKL$.

The ratio of the sides of a parallelogram is equal to the ratio of its heights, as can be seen by drawing the two heights from one corner and noting the two similar triangles that are formed.

We can find the ratio of the heights in $QRST$ by drawing a normal AF from A to the (extension of) side ST and a normal BG from B to the (extension of) side RS . The triangles AFC and BGD are similar, because $FC \perp BD$ and $AC \perp GD$. Thus, $BG : AF = BD : AC$.

We have now shown that $QRST$ and $LMNK$ are parallelograms with the same angles and the same side ratios, so they are similar.